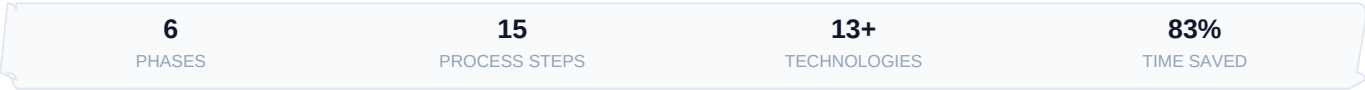


# AI Inventory Intelligence Pipeline

End-to-end process map: from fragmented data sources to AI-generated insights, automated dashboards, and action items across 8+ retail locations. Built as Product Management Supervisor at Canada's Wonderland, 2025.



## PHASE 1 Data Collection

- 1

### POS transaction export

Daily sales and transaction data exported from DataWorks POS system. Covers all 8+ retail locations: Thrills, Alpine Outfitters, Main Gate, and others. Includes SKU, quantity, price, location, timestamp.

IN: Real-time POS transactions  
OUT: Daily CSV transaction exports per location

DataWorks POS

CSV Export
- 2

### Inventory file collection

Stock counts, receiving logs, inter-store transfer records, and shrink reports pulled from Excel-based inventory tracking system maintained per location.

IN: Physical counts + system records  
OUT: Excel inventory files per location

Microsoft Excel

Manual Counts
- 3

### Location performance snapshots

Weekly location-level data: revenue, guest traffic, staffing hours, shrink rate, top/bottom sellers, and operational notes from store supervisors.

IN: Store manager weekly inputs  
OUT: Location KPI snapshots

DataWorks

Excel

Location Managers

## PHASE 2 Data Pipeline & Processing

- 1

### Data ingestion

Python script reads all source files: DataWorks CSV exports, Excel inventory sheets (openpyxl), and location report files. Handles varying schemas and formats across locations.

IN: Multiple CSV/Excel files from Phase 1  
OUT: Raw dataframes loaded into memory

Python

pandas

openpyxl

csv
- 2

### Cleaning & standardization

Normalize SKU names and codes, fix data types, handle missing/null values, remove duplicate transactions, reconcile discrepancies between POS counts and physical inventory.

IN: Raw dataframes  
OUT: Clean, standardized dataset

Python

pandas
- 3

### Merging & metric calculation

Join transaction data with inventory data on SKU + location. Calculate derived KPIs: sell-through rate, days-of-supply, inventory turnover ratio, shrink percentage, revenue per sq ft.

IN: Clean POS + inventory data  
OUT: Unified dataset with 15+ calculated KPIs

Python

pandas

NumPy

## PHASE 3

## AI Agent Processing

1

**Copilot report generation**

Microsoft Copilot agent connected via Office 365 connectors. Receives weekly data summary. Generates natural language performance report per location, auto-flags anomalies (shrink spikes, unusual sales patterns, inventory discrepancies).

Microsoft Copilot

Office 365

Excel

Connectors

IN: Unified dataset summary tables  
OUT: Drafted weekly report + anomaly flags

2

**ChatGPT SKU analysis**

Python script sends cleaned dataset to GPT-4 via OpenAI API. Prompt instructs: analyze SKU-level sell-through, identify slow movers for markdown, flag stockout risks, cross-reference assortment gaps between locations (e.g., Alpen Fury merch in Thrills vs Alpine Outfitters).

ChatGPT / GPT-4

OpenAI API

Python

Prompt Engineering

IN: Unified dataset + location SKU breakdown  
OUT: JSON: reorder points, markdown candidates, ga...

3

**Recommendation validation**

Python script validates AI outputs: checks reorder quantities against budget constraints, verifies SKU IDs exist in master catalog, ensures markdown percentages within policy limits (max 40%).

IN: AI-generated recommendations  
OUT: Validated, filtered recommendations

**DECISION:** Valid: Proceed to routing | Invalid: Flag for manual review



## PHASE 4

## Dashboard &amp; KPI Tracking

1

**KPI dashboard update**

Python script writes processed data and calculated KPIs to Excel/DataWorks dashboard. 15+ metrics: sell-through, turnover, stockouts, shrink, revenue/location, top/bottom SKUs, days-of-supply, replenishment rate, markdown effectiveness.

Excel

DataWorks

Python

openpyxl

IN: Unified dataset + calculated metrics  
OUT: Updated dashboard, 15+ KPIs, 8+ locations

2

**Anomaly highlighting**

Conditional formatting applied programmatically: red for stockout risk, amber for slow movers, green for healthy metrics. Thresholds defined per KPI (e.g., turnover < 1.5x = red, shrink > 3% = red).

IN: KPI values + threshold rules  
OUT: Color-coded dashboard with flagged items

**DECISION:** Breach threshold: Red/Amber flag | Normal: Green



## PHASE 5

## Action Routing &amp; Alerts

1

**Slack alerts**

Automated Slack messages to #ops and #leadership channels: weekly performance summary, critical anomalies (shrink spikes, stockout warnings), and top-line numbers requiring attention.

Slack API

Python

Webhook

IN: Report summary + anomaly flags  
OUT: Slack notifications delivered

